

# SYDNEY TECHNICAL HIGH SCHOOL

## YEAR 12 ASSESSMENT TASK 3

**JUNE 2004**

### **MATHEMATICS: 2 UNIT**

Time Allowed: 70 minutes

Instructions:

- Write your name and class at the top of this page, and at the top of each answer sheet
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions are of equal value. Attempt all questions.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Standard integrals are attached at the back of this paper.

Name: .....

Teacher: .....

| Q1  | Q2  | Q3  | Q4  | Q5  | Total |
|-----|-----|-----|-----|-----|-------|
| /12 | /12 | /12 | /12 | /12 | /60   |

### Question 1 (12 Marks)

a) For the parabola  $x^2 - 4x - 8y - 36 = 0$

find : i) the co-ordinates of the vertex (1)

ii) the focal length (1)

iii) the co-ordinates of the focus (1)

iv) the equation of the directrix (1)

b) If  $x^{1.0986} = 3$  find:

i)  $\log_x 3$  (1)

ii)  $\log_x 81$  (1)

c) Evaluate  $\int_0^{\pi/2} \sin 2x \, dx$  (2)

d) Differentiate  $y = x \tan x$  (2)

e) Differentiate  $y = \log_e(4 - x^2)$  (2)

### Question 2

a) Sketch the graph of  $y = \cos \frac{x}{4}$  for  $-4\pi \leq x \leq 4\pi$  (2)

b) (i) Evaluate  $\int_0^{2\pi} \cos \frac{x}{4} dx$  (2)

(ii) For the function in (i)  $y = \cos \frac{x}{4}$ , complete the table below

|     |   |                 |       |                  |        |
|-----|---|-----------------|-------|------------------|--------|
| $x$ | 0 | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ | $2\pi$ |
| $y$ |   |                 |       |                  |        |

Hence approximate  $\int_0^{2\pi} \cos \frac{x}{4} dx$  using Simpson's rule with 5 function values and answering correct to 2 decimal places. (3)

### Question 2 (cont)

(iii) Find the percentage error given by using Simpson's rule to approximate the integral (answer correct to 1 decimal place). (1)

- c) (i) Find the co-ordinates of the point on the curve  $y = x \ln x$  for which the gradient of the tangent is equal to 2. (2)
- (ii) Hence find the equation of the normal at this point. (2)

### Question 3

a) Solve the equation  $5e^{2x} = 0.4213$ . Give your answer to 4 decimal places. (2)

b) By using the substitution  $u = e^x$  or otherwise, solve the equation  $e^{2x} - 6e^x + 5 = 0$   
Answer to 3 decimal places where necessary (3)

c) Solve for  $\theta$  in the domain  $0 \leq \theta \leq 2\pi$ : (3)

$$4 \sin \theta - 3 \cos \theta = 0$$

d) Differentiate (2)

$$y = \sqrt{\sin 2x}$$

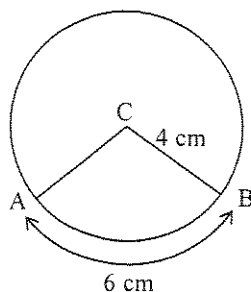
e) Find the derivative of  $y = e^{\cos x}$  (1)

### Question 4

- a) Find
- i)  $\int e^{5x} dx$  (1)
- ii)  $\int \sqrt{e^x} dx$  (2)
- b) Express  $\frac{2\pi^c}{5}$  in degrees (1)

#### Question 4 (cont.)

c)



In the diagram, AB is an arc of length 6cm in a circle, centre C, radius 4cm. Find the

- i) size of angle ACB in radians (1)
- ii) area of sector ACB (1)
- iii) area of the minor segment cut off by the chord AB (2)

- d) (i) Find the area bounded by the  $x$  axis, the curve  $y = e^x$  and the lines

$$x = 0 \text{ \& } x = \log_e 5. \quad (2)$$

- (ii) This area is rotated about the  $x$  axis.

Find the volume of the solid so formed. (Leave answer in terms of  $\pi$ ). (2)

#### Question 5

a) Solve for  $x$ ,  $\sin^2 2x = \frac{1}{4}$  ( $0 \leq x \leq 180^\circ$ ) (2)

- b) (i) Find the value of  $k$  for which the equation  $x^2 - (k-5)x + (k-7) = 0$  has roots which are reciprocals of one another. (1)

- (ii) If  $\alpha$  and  $\beta$  are the roots of this equation find

A)  $\alpha + \beta$  (1)

B)  $\frac{1}{\alpha} + \frac{1}{\beta}$  (1)

c) Find the values of  $P, Q$  and  $R$  if  $3x^2 + 5x - 1 \equiv P(x+1)^2 + Q(x+1) + R$ . (3)

d) Find all the values of  $k$  for which  $(k+2)x^2 + x - 3 = 0$  has real roots. (2)

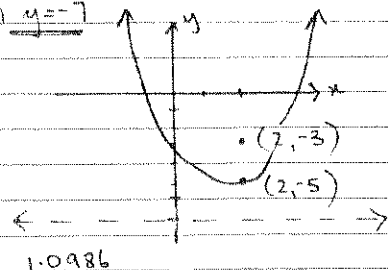
e) Evaluate  $\int_1^2 \frac{x^3 - 3x + 2}{x^2} dx$  (2)

End of Exam.

### Question 1

a)  $x^2 - 4x + 4 = 8y + 36 + 4$   
 $(x-2)^2 = 8y + 40$   
 $(x-2)^2 = 8(y+5)$

- i) Vertex (2, -5) ①  
 ii)  $4a = 8 \therefore a = 2$  ①  
 iii) Focus (2, -3) ①  
 iv)  $y = -7$  ①



b)  $x = 3$   
 i)  $\log_3 3 = 1.0986$  ①  
 ii)  $\log_8 1 = \log_3 3$   
 $= 4 \log_3 3$   
 $= 4.3944$  ①

c)  $\int_0^{\pi/3} \sin 2x dx = \left[ -\frac{1}{2} \cos 2x \right]_0^{\pi/3}$   
 $= -\frac{1}{2} \left[ \cos 2\pi/3 - \cos 0 \right]$   
 $= -\frac{1}{2} \left[ -\frac{1}{2} - 1 \right]$   
 $= 3/4$  ①

d)  $\frac{d}{dx} (x \tan x)$

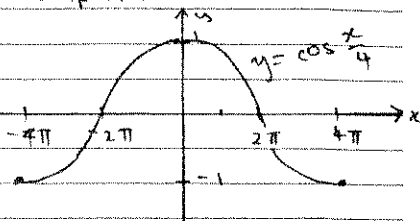
$u = x \quad v = \tan x$   
 $u' = 1 \quad v' = \sec^2 x$   
 $\frac{dy}{dx} = \tan x + x \sec^2 x$  ②

e)  $\frac{d}{dx} \ln(4-x^2) = \frac{-2x}{4-x^2}$  ②

### Question 2

a) period =  $\frac{2\pi}{1/4} = 8\pi$

amplitude = 1



b) i)  $\int_0^{2\pi} \cos \frac{x}{4} dx$   
 $= \left[ 4 \sin \frac{x}{4} \right]_0^{2\pi}$   
 $= 4 \left( \sin \frac{\pi}{2} - \sin 0 \right)$   
 $= 4$

ii)

|   | F | L       |
|---|---|---------|
| x | 0 | $\pi/2$ |
| y | 1 | 0       |

$y_1 = 1, y_2 = 0$

$S = \frac{\pi/2}{3} \left[ 1 + 0 + 4(1 + 0) \right]$   
 $= 4.00$

c) i)  $y = x \ln x \therefore \frac{dy}{dx} = 2$

$u = x \quad v = \ln x$   
 $u' = 1 \quad v' = \frac{1}{x}$   
 $\frac{dy}{dx} = \ln x + 1$

$\ln x + 1 = 2$   
 $\ln x = 1$

$\log_e x = 1 \therefore e = x$

$\therefore \text{pt } (e, e)$

ii)  $m = -\frac{1}{2}$  at (e, e)

eqn  $y - e = -\frac{1}{2}(x - e)$

$2y - 2e = -x + e$

$x + 2y - 3e = 0$

### Question 3

a)  $5e^{2x} = 1.4213$   
 $e^{2x} = \frac{1.4213}{5}$

$2x \ln e = \ln(0.28426)$   
 $x = -1.2369$  (4 d.p.)

b)  $e^{2x} - 6e^x + 5 = 0$   
 $u^2 - 6u + 5 = 0$   
 $(u - 5)(u - 1) = 0$   
 $u = 5 \quad u = 1$   
 $\therefore x = 1.609 \quad x = 0$

c)  $4 \sin \theta - 3 \cos \theta = 0$   
 $4 \sin \theta = 3 \cos \theta$   
 $\tan \theta = \frac{3}{4}$   
 $\theta = 0.64^\circ, 3.79^\circ$

d)  $y = (\sin 2x)^{1/2}$   
 $\frac{dy}{dx} = \frac{1}{2} (2 \cos 2x) (\sin 2x)^{-1/2}$   
 $= \frac{\cos 2x}{\sqrt{\sin 2x}}$

e)  $\frac{d}{dx} (e^{\cos x}) = -\sin x \cdot e^{\cos x}$

### Question 4

a) i)  $\int e^{5x} dx = \frac{1}{5} e^{5x} + C$

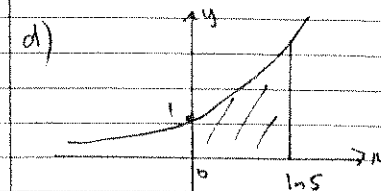
ii)  $\int e^{x/2} dx = 2e^{x/2} + C$   
 OR  $2\sqrt{e^x} + C$

b)  $\frac{2\pi}{5} = 72^\circ$

c) i)  $b = 4 \times \theta$   
 $\theta = 1.5^\circ$   
 $\therefore \angle ACB = 1.5^\circ$

ii)  $A = \frac{1}{2} \times 4 \times 1.5$   
 $= 12 \text{ cm}^2$

iii)  $A = \frac{1}{2} \times 4^2 (1.5 - \sin 1.5)$   
 $= 4.02 \text{ cm}^2$

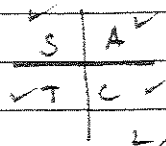


d) i)  $A = \int_0^{\ln 5} e^x dx$   
 $= \left[ e^x \right]_0^{\ln 5}$   
 $= e^{\ln 5} - e^0$   
 $= 4 \text{ u}^2$

ii)  $V = \pi \int_0^{\ln 5} e^{2x} dx$   
 $= \pi \left[ \frac{e^{2x}}{2} \right]_0^{\ln 5}$   
 $= \frac{\pi}{2} [e^{2 \ln 5} - e^0]$   
 $= \frac{\pi}{2} [25 - 1]$   
 $= 12\pi \text{ unit}^3$

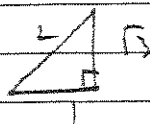
# Question 5

a)  $\sin^2 2x = \frac{1}{4}$



$\sin 2x = \pm \frac{1}{2}$

acute  $2x = 30^\circ$



$\therefore 2x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

$x = 15^\circ, 75^\circ, 105^\circ, 165^\circ$

b) i) Reciprocal roots  $\alpha, \frac{1}{\alpha}$

$\therefore$  product is 1  $\therefore \frac{c}{a} = 1$

$k-7 = 1$

1

$k = 8$

ii) A)  $\alpha + \beta = \frac{k-5}{1} = \underline{3}$

B)  $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \underline{\underline{\frac{3}{1}}}$

c)  $3x^2 + 5x - 1 = P(x+1)^2 + Q(x+1) + R$   
 $= P(x^2 + 2x + 1) + Qx + Q + R$   
 $= Px^2 + 2Px + P + Qx + Q + R$

$= P(x^2) + x(2P + Q) + P + Q + R$

$\therefore P = 3$

$2P + Q = 5$

$3 - 1 + R = -1$

$6 + Q = 5$

$R = -3$

$Q = -1$

d) Real Roots  $\Delta \geq 0$

$1 - 4(k+2)x - 3 \geq 0$

$1 + 12(k+2) \geq 0$

$1 + 12k + 24 \geq 0$

$12k \geq -25$

$k \geq \frac{-25}{12}$

e)  $\int_1^2 \frac{3x^2 - 3x + 2}{x^2} dx = \int_1^2 x - \frac{3}{x} + 2x^{-2} dx$

$= \left[ \frac{x^2}{2} - 3 \ln x - \frac{2}{x} \right]_1^2$

$= \left[ (2 - 3 \ln 2 - 1) - \left( \frac{1}{2} - 0 - 2 \right) \right] = \underline{\underline{2\frac{1}{2} - 3 \ln 2}}$